Thomas Moerman

PhD researcher in natural language processing and AI, Ghent University (LT3)

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My research develops retrieval-augmented and synthetic-data methods for language technology, with machine translation as the primary focus and applications in educational NLP, fairness in language models, and vision-language OCR. A consistent finding is that these techniques let smaller specialized models reach or exceed the quality of much larger general-purpose LLMs.

EDUCATION

PhD in Natural Language Processing, Ghent University, LT3 — 2023 to 2027 (expected)
Machine translation augmented with automatically extracted similar translations. Supervisors: Arda Tezcan and Els Lefever.

Advanced MA in Linguistics: Natural Language Processing, Ghent University and KU Leuven, summa cum laude — 2023

MA in Linguistics and Literature, English and Scandinavian studies, Ghent University, cum laude — 2022

BA in Linguistics and Literature, English and Swedish, Ghent University, cum laude — 2021

SELECTED PUBLICATIONS

9 published or accepted works, 6 first-authored, 3 journal articles. Full list: Google Scholar and ORCID

Moerman, Gkovedarou, Hackenbuchner. **ContraGAND: Auditing and Repairing Gender Ambiguity Failures in LLMs with Neurosymbolic Contrastive Data Augmentation.** EMNLP, main conference. — 2026

Moerman, Tezcan, Macken. **Multilingual Communication in the Asylum Context: LLM-Based MT with Fuzzy Match Augmentation.** EAMT. Shortlisted for the best-paper award. — 2026

Moerman, Degraeuwe, Tezcan. **Fuzzy Semantic Retrieval Strategies for Automated Short-Answer Grading with LLMs.** Computational Linguistics in the Netherlands Journal, 15. — 2026

Moerman, Tezcan. **Getting More from Small LLMs: Fuzzy-Match Retrieval and Candidate Scoring for Multitask Sorbian NLP.** WMT shared task system description, under review. — 2026

Moerman, Lefever, Tezcan. **Advancing Fuzzy Match Augmentation for Domain-Specific Machine Translation.** Under review at the Journal of Artificial Intelligence Research. — 2025

AWARDS AND COMPETITIONS

Winner, Sorbian track, WMT 2026 shared task on multitask LLMs with limited resources (team LT3, team lead). One 2B model jointly handling translation, QA, spelling, grammar, and maths tasks for Upper and Lower Sorbian. — 2026

Best-paper award shortlist, EAMT 2026. — 2026

RESEARCH GRANTS

Principal applicant, VSC Tier-1 compute project grant on the Sofia cluster (38,546 GPU hours), after earlier Tier-1 project allocations (2024, 2025) and a Tier-1 Starting Grant (2024). — 2024 to 2026

Named collaborator on five further Tier-1 compute grants across four Ghent University research groups, contributing LLM fine-tuning and retrieval pipelines. — 2025 to 2026

TEACHING

Guest lectures on Neural Machine Translation and on LLMs in Translation (MTPE course, Ghent University). Co-taught Introduction to Python for Humanities. Hands-on HPC workshop for the LT3 team. Python module at the MILS summer school (Methods in Language Sciences).

ACADEMIC SERVICE

Reviewer for ACL, EAMT, and MT Summit. Presented at EAMT 2026, MT Summit 2025, the Machine Translation Marathon 2025, CLIN, and LREC 2026.

TECHNICAL SKILLS AND LANGUAGES

Python, PyTorch, Hugging Face Transformers, vLLM, axolotl, LoRA fine-tuning, FAISS retrieval, OpenNMT, Moses, SLURM on Tier-1 HPC (A100 and H200 clusters).
Dutch (native), English (C2), Swedish (C1), French (B2), German (B1), Norwegian (B1).